

Protein Allergen Classification System

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Abstract—The purpose of this project is to establish an end-to-end machine learning framework to identify if a protein sequence is an allergen and classify it into one of the six allergen categories. The World Allergy Organization states allergic diseases are becoming a global epidemic affecting 30% of the world population. The World Health Organization has stated the allergenic potential of a new protein in therapeutic agents or modified food crops is an emerging issue that requires new methods and safety assessment of therapeutics.

The pipeline contains two working steps. Step A, the binary classifier, identifies allergens as opposed to proteins using an XGBoost model that classifies allergens among non-allergens based on a set of 64 physicochemical and dipeptide features. This model has an accuracy of 96.1% with a ROC-AUC of 0.994 on a held-out test set. Step B, the multi-class classifier, classifies allergens into one of six classes (Food, Pollen, Venom, Dust Mite, Latex, or Mold/Fungal) using a Random Forest model with a F1 score of 0.944.

Direct protein sequence retrieval from NCBI's Entrez API is followed by real-time feature extraction and allergen classification through a Flask-based web application. All components of the model are deployed as a single `trained_model.pkl` file.

Index Terms—allergen classification, machine learning, protein sequences, XGBoost, Random Forest, bioinformatics

I. INTRODUCTION

An allergy is when the body's immune system incorrectly recognizes certain proteins as dangerous, causing it to overreact. These proteins can be present in food, pollen, insect venom, dust mites, latex, and mold, and symptoms can range from an uncomfortable reaction to a life-threatening anaphylactic reaction. In order to mitigate an allergic reaction, it is of utmost importance to determine which proteins a person is allergic to. This is especially important in the safety regulations of food products, pharmacology, and the assessment of environmental health.

The traditional methods of assessing allergenicity include structural similarity searches among certain allergen databases (AllerTop, AllerHunter, etc.) or tests that provide direct immunological reactions in a lab. These methods are not perfect, though. Searching through a database can overlook allergens that are structurally divergent, and the lab tests can be time-consuming and costly.

In the context of allergenicity, machine learning (ML) can play an important role. Certain types of ML are capable of classifying proteins beyond the traditional similarity methods.

This study aims to refine and present a machine learning approach to allergenicity classification by providing the following sets of data and features:

- 1) **Large, balanced dataset.** 15,000 allergen sequences (2,500 for each allergen class) and 15,000 non-allergen sequences obtained from NCBI, totaling nearly 30,000 protein records.
- 2) **Two-stage classification.** This method is meant to classify allergens at two different levels. The first stage uses a binary model to determine if a given protein is an allergen. The second stage is meant to classify the determined allergen into one of the several classes of allergens.
- 3) **Rich feature engineering.** This method includes the calculation of 64 different parameters that describe allergenic proteins and potentially non-allergenic proteins.
- 4) **A system that is fully implemented and operational.** The models created in this paper have been made available through a REST API (Flask) and a purpose-built HTML/JS interface.

II. DATA COLLECTION

A. Allergen Sequences

The Protein sequences were extracted using the Biopython Entrez API from the NCBI Protein data repository. To make sure that all allergen categories are equally represented, each of the six allergen categories was queried independently.

TABLE I: Allergen categories, their sources, and example proteins.

Category	Primary Sources	Named Codes (IUIS)
Food Allergen	Peanut, soy, wheat, milk, egg, fish, shellfish	Ara h, Gly m, Tri a, Bos d
Pollen Allergen	Birch, grass, ragweed, olive, cypress	Bet v, Phl p, Amb a, Ole e
Venom Allergen	Bee, wasp, ant venoms	Api m, Ves v, Sol i
Dust Mite Allergen	<i>Dermatophagoides</i> spp., storage mites	Der p, Der f, Blo t
Latex Allergen	<i>Hevea brasiliensis</i> (rubber tree)	Hev b
Mold/Fungal Allergen	<i>Alternaria</i> , <i>Aspergillus</i> , <i>Cladosporium</i>	Alt a, Asp f, Cla h
Non-Allergen	Reviewed housekeeping, structural, enzyme proteins	—

B. ID Collection Strategy

A lazy-fetch strategy was implemented. IDs were collected query-by-query until a pool was reached that was $1.4 \times$ target (i.e. 3,500 for each allergen category). This 40% buffer accommodates for a loss of sequences that would not make the quality cut, without the ID overload that many prior approaches have incorporated.

C. Quality Filtering

Sequences were accepted if they:

- Had length ≥ 30 amino acids.
- Contained only standard amino acid characters (ACDEFGHIKLMNPQRSTVWY).
- Were unique (de-duplicated by accession ID).

D. Final Dataset

TABLE II: Final dataset composition.

Class	Rows	Fraction
Food Allergen	2,500	12.5%
Pollen Allergen	2,500	12.5%
Venom Allergen	2,500	12.5%
Dust Mite Allergen	2,500	12.5%
Mold/Fungal Allergen	2,500	12.5%
Latex Allergen	$\leq 2,500$	$\leq 12.5\%$
Non-Allergen	15,000	25.0%
Total	$\approx 30,000$	100%

III. EXPLORATORY DATA ANALYSIS

A. Class Distribution

Generally, in the allergen type distributions, most samples showed non-allergens having the biggest group, with about 15,000 members. The other six types had about 2,500 members each. Latex allergens were under-represented due to the low quality sequences for the *Hevea brasiliensis* in the NCBI database.

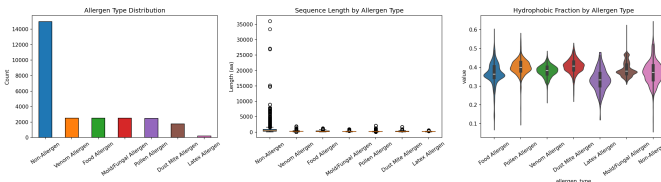


Fig. 1: EDA overview. **Left:** Allergen type distribution. **Centre:** Sequence length by allergen type. **Right:** Hydrophobic fraction by allergen type.

B. Sequence Length

Most allergen sequences ranged from short to medium length (50–600 residues), while some long non-allergen sequences extended to 35,000 residues and included structural proteins and enzymes. Therefore, sequence length is an important discriminating factor.

C. Physicochemical Properties

Many of the allergen types fell within the 0.35–0.40 range for the hydrophobic fraction. Hence, when it comes to type discrimination, hydrophobicity is of limited value. More informative differences are visible in the per-type boxplots of Figure 2.

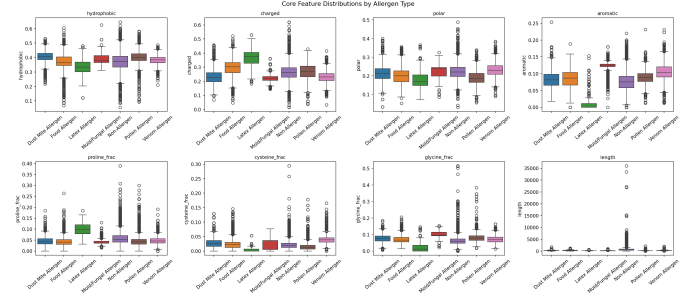


Fig. 2: Core feature distributions by allergen type. Key discriminating patterns: Latex allergens have very low cysteine and aromatic fractions but a very high glutamic acid (aa_E) fraction. Mold/Fungal allergens show the highest aromatic fraction.

Some unique patterns for each type included the following:

- **Latex Allergens:** Extremely high glutamic acid (aa_E ≈ 0.23) and extremely low cysteine fraction (≈ 0.004), due to the *Hevea* protein composition.
- **Mold/Fungal Allergens:** The highest aromatic fraction (aa_F, aa_Y, aa_W combined ≈ 0.12) and a short average length of about 162 amino acids.
- **Venom Allergens:** Elevated cysteine fraction (≈ 0.039), indicating toxin structures that are disulphide stabilised.
- **Non-Allergens:** A highly varied and dissimilar group, as expected for a heterogeneous class.

IV. FEATURE ENGINEERING

A. Feature Types

Each protein structure is encoded as 64 numerical features. This 3-part encoding includes:

- 1) **Physicochemical group fractions** (8 features): fractional representations of hydrophobic, charged, polar, and aromatic groups, as well as proline, cysteine, glycine fractions, and sequence length.
- 2) **Per-amino-acid frequencies** (20 features): fractional representations of the 20 standard amino acids.
- 3) **Dipeptide frequencies** (selected subset): normalised frequencies of selected two-residue subsequences.

B. Mathematical Definitions

Let s be a protein sequence of length n over the standard amino acid alphabet $\mathcal{A} = \{A, C, D, \dots, Y\}$.

Amino acid frequency:

$$f_a = \frac{|\{i : s_i = a\}|}{n}, \quad a \in \mathcal{A} \quad (1)$$

Group fraction (e.g., hydrophobic $\mathcal{H} = \{A, V, I, L, M, F, Y, W\}$):

$$f_{\mathcal{H}} = \sum_{a \in \mathcal{H}} f_a \quad (2)$$

Dipeptide frequency:

$$d_{ab} = \frac{|\{i : s_i = a, s_{i+1} = b\}|}{n-1}, \quad a, b \in \mathcal{A} \quad (3)$$

C. Feature Selection

After variance and pairwise correlation filtering at $|r| > 0.88$, raw features are scored using a weighted sum of four methods as follows:

$$\text{score}_j = 0.30 \cdot \tilde{I}_j^{\text{RF}} + 0.30 \cdot \tilde{I}_j^{\text{ET}} + 0.25 \cdot \tilde{I}_j^{\text{MI}} + 0.15 \cdot \tilde{F}_j \quad (4)$$

where $\tilde{I}^{\text{RF}}$, $\tilde{I}^{\text{ET}}$, $\tilde{I}^{\text{MI}}$, and $\tilde{F}$ are min-max normalised Random Forest importances, Extra Trees importances, mutual information scores, and ANOVA F -statistics, respectively. This scoring was performed independently for Stage A (binary) and Stage B (6-class allergen-type), and the union of the two selected sets was retained.

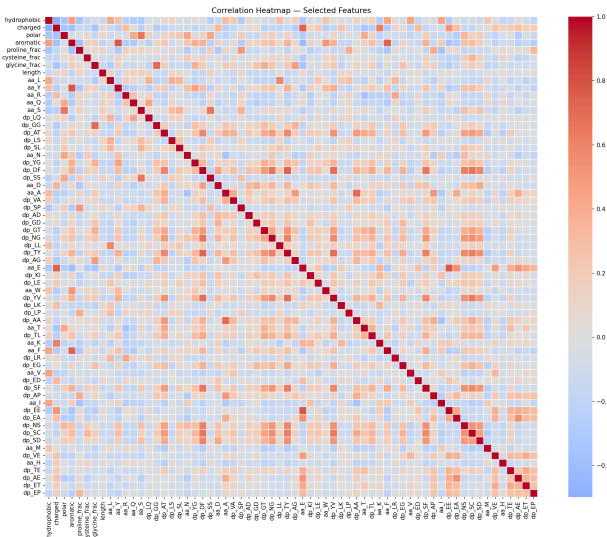
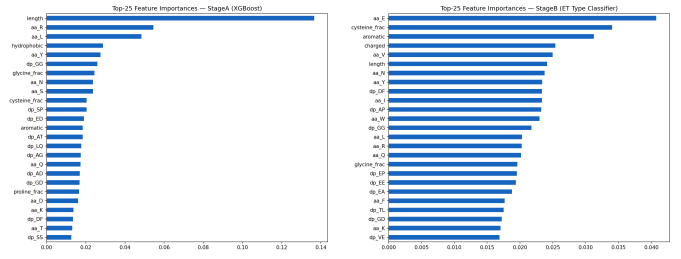


Fig. 3: Correlation heatmap of selected features. Most feature pairs are weakly correlated (near-zero, light blue/white), confirming that the correlation filter removed redundant dipeptide pairs effectively. Strong correlations remain only on the diagonal.



(a) Stage A (XGBoost): top-25 features for binary allergen detection. Sequence length dominates. (b) Stage B (RF): top-25 features for allergen type. Glutamic acid (aa_E) and cysteine fraction lead.

Fig. 4: Feature importance rankings for Stage A and Stage B models.

The top features for each stage differ considerably:

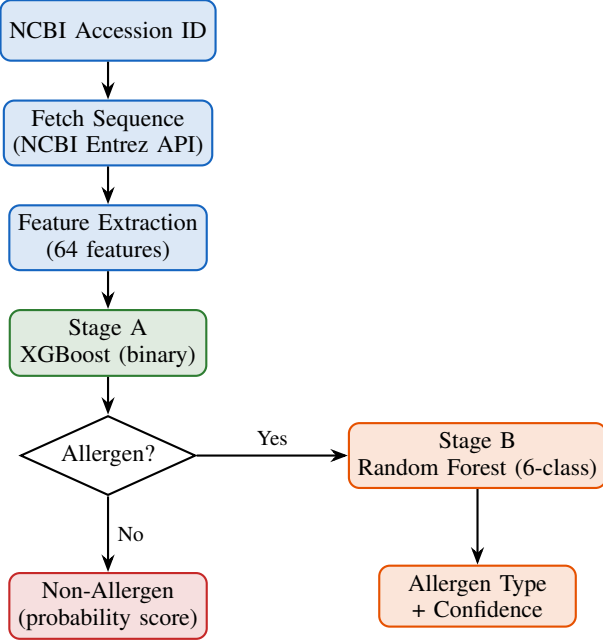
- **Stage A:** Sequence length has by far the highest importance (0.135), followed by aa_R (arginine frequency) and aa_L (leucine frequency). This reflects the fact that allergens tend to be globular proteins of moderate, consistent length.
- **Stage B:** aa_E (glutamic acid, 0.040) and cysteine_frac (0.034) lead, driven by the extreme glutamic acid enrichment of Latex allergens and the elevated cysteine content of Venom allergens.

V. MODEL ARCHITECTURE

A. Two-Stage Design Rationale

Taking the approach of putting together a single multi-class model in service of all seven classes (six types of allergens + non-allergen) led to an overall poor model. This can be attributed to the fact that in allergen versus non-allergen cases, the features that serve to distinguish these cases vary from those used in non-allergen versus varieties of allergens. Binary modelling in Stage A and Stage B for within-class differentiation of allergens allows for the construction of models tailored to each task and provides better performance for each task.

B. System Workflow



C. Stage A — Binary Classifier

Model: XGBoost (XGBClassifier) with 300 trees, learning rate 0.08, max depth 6, subsample 0.8, column subsampling 0.8, and `scale_pos_weight` set to the non-allergen/allergen ratio (≈ 0.33) to handle the 3:1 allergen/non-allergen imbalance.

Three candidate models were evaluated on the validation set:

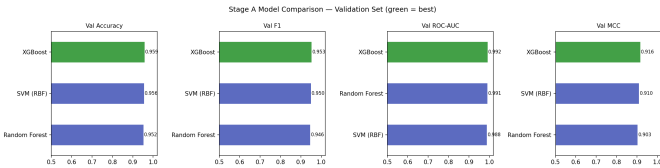


Fig. 5: Stage A model comparison on the validation set. XGBoost (green) outperforms SVM (RBF) and Random Forest on all four metrics.

Among the three models tested, XGBoost achieved the best validation F1 (0.953) and ROC-AUC (0.992), and was re-trained on the combined train+validation set before final evaluation on the held-out test set.

D. Stage B — Allergen Type Classifier

Model: Random Forest classifier with balanced class weights, trained exclusively on allergen-only rows.

Note on rule-based approaches: The first version of this system incorporated hand-crafted thresholds (e.g. “charged > 0.18 $\Rightarrow$ Pollen Allergen”). This resulted in type accuracy of around 13%, since thresholds like “charged > 0.18” are met by over 90% of all allergen types. In comparison, the trained Random Forest model reaches 94.6% type accuracy.

Three models for Stage B were assessed:

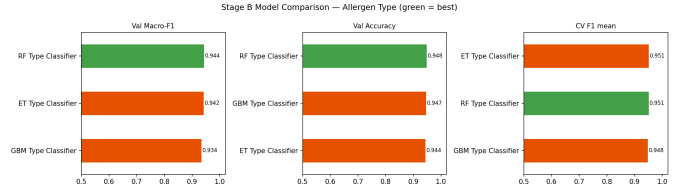


Fig. 6: Stage B model comparison. Random Forest (green) achieves the best macro-F1 (0.944) and is chosen for the final system.

Based on the best macro-F1 (0.944), Random Forest was selected for the final system.

E. Train / Validation / Test Split

A stratified split of 70% training, 15% validation, and 15% testing was executed on the 7-class label (6 allergen types and Non-Allergen) to ensure that each class is represented in each of the splits in proportions relative to the overall dataset.

$$\text{Train : } 70\% \approx 14,000 \text{ samples,} \quad (5)$$

$$\text{Val : } 15\% \approx 3,000 \text{ samples,} \quad (6)$$

$$\text{Test : } 15\% \approx 3,000 \text{ samples} \quad (7)$$

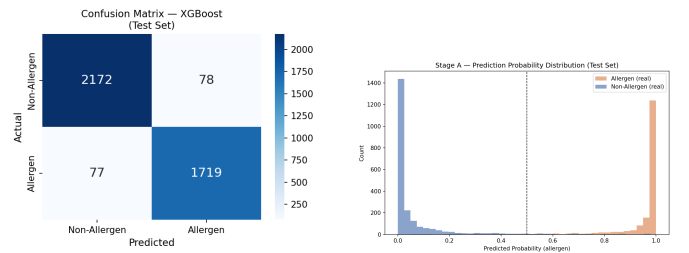
VI. RESULTS

A. Stage A — Binary Classification (Test Set)

TABLE III: Stage A (XGBoost) — final test set metrics.

Metric	Value
Accuracy	0.961
F1 Score (binary)	0.953
ROC-AUC	0.9940
Average Precision	0.9930
MCC	0.916
TP / TN / FP / FN	1719 / 2172 / 78 / 77

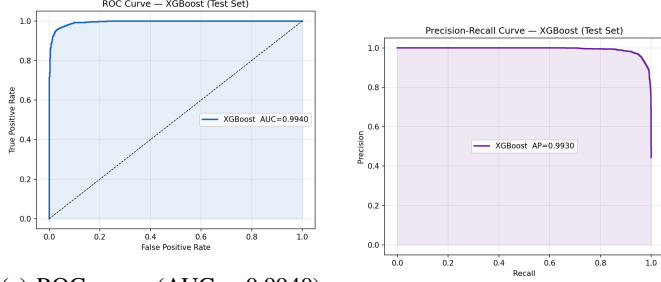
a) Confusion Matrix.: The model made a total of 155 errors out of 4,046 samples (78 false positives, 77 false negatives). The errors within the confusion matrix show the model is not skewed towards predicting positive or negative samples.



(a) Stage A confusion matrix (XGBoost, test set). (b) Predicted probability distribution by true class.

Fig. 7: Stage A classification results.

b) Probability Distribution.: Near-perfect separation in the probability distribution is observed. This means true positives are near probability 1.0 and negative samples are near probability 0.0, with very little overlap in the 0.2–0.8 range. The model is highly accurate and confidence scores are meaningful and well-calibrated.



(a) ROC curve (AUC = 0.9940). The curve hugs the top-left corner, indicating near-perfect discrimination. (b) Precision-Recall curve (AP = 0.9930). Precision remains near 1.0 until recall ≈ 0.97 , then drops.

Fig. 8: ROC and PR curves for Stage A on the test set.

c) ROC and Precision-Recall Curves.: The ROC score and average precision support that even at high levels of recall, the accuracy of the model would remain high. This is of great importance given the context of ensuring safety in allergen testing, where a missing true positive could have negative effects.

B. Stage B — Allergen Type Classification (Test Set)

TABLE IV: Stage B (Random Forest) — allergen type classification on allergen-only test rows.

Allergen Type	Precision	Recall
Dust Mite Allergen	0.934	0.928
Food Allergen	0.969	0.971
Latex Allergen	1.000	0.968
Mold/Fungal Allergen	0.997	0.997
Pollen Allergen	0.980	0.969
Venom Allergen	0.956	0.978
Macro Avg	0.973	0.969

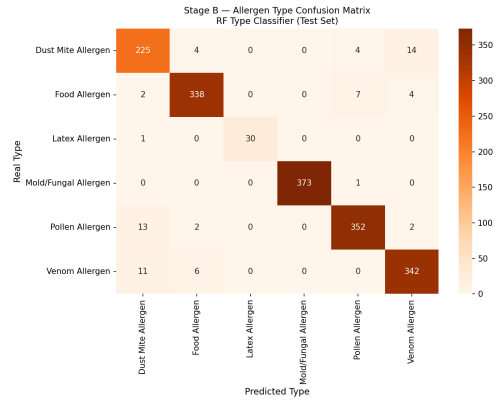


Fig. 9: Stage B allergen-type confusion matrix (Random Forest Classifier, test set). The strong diagonal confirms high per-class accuracy. The most common errors are Dust Mite $\leftrightarrow$ Pollen and Dust Mite $\leftrightarrow$ Venom, which is expected given their overlapping physicochemical profiles.

a) Confusion Matrix.: Off-diagonal errors identify most Pollen and Venom allergen errors (11 and 13 errors respectively), caused by overlap with Dust Mite allergens. Latex allergens are well-defined due to their unique structure and therefore yielded 30 out of 31 correct answers with no false positives.

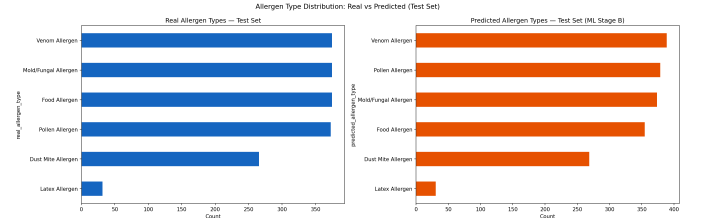


Fig. 10: Real vs. predicted allergen type counts on the test set. The predicted distribution closely mirrors the real distribution across all six types, confirming that the model is not systematically biased toward any particular category.

b) Prediction Distribution.: The model predicts the distribution of the different allergen types without bias and accurately across all six types.

VII. SYSTEM ARCHITECTURE & DEPLOYMENT

A. Backend (Flask)

The Flask backend (`app.py`) deploys the `trained_model.pkl` file as a bundle and offers a single REST endpoint:

```
POST /predict
Body: { "sequence": "<amino acid string>" }
Response: {
  "is_allergen": true | false,
  "stage_a_probability": 0.87,
  "allergen_type": "Pollen Allergen" | null,
  "stage_b_probabilities": {
```

```

    "Food Allergen": 0.05, ...
} | null
}

```

The backend performs the same feature extraction (64 features) used in training, applies Stage A, and conditionally proceeds to Stage B.

B. Frontend (HTML/JS)

Displayed in the browser UI (`allergen_ui.html`) are:

- Input fields for NCBI protein accession IDs.
- A live sequence fetch from NCBI's Entrez eFetch API.
- A confidence ring (dynamic SVG) depicting Stage A probability.
- A classification of Stage B type with probability bars of each class.
- A bar chart of physicochemical features.
- A sequence viewer with colour-coded cysteine residues.

C. Model Bundle

The `trained_model.pkl` file contains:

- `stage_a_model`: XGBoost binary classifier (allergen vs. non-allergen).
- `stage_b_model`: Random Forest 6-class allergen-type classifier.
- `label_encoder_6`: Maps integer predictions to allergen type strings.
- `feature_names`: Ordered list of 64 feature names (exact column order).
- Assessment metrics with documentation added by the author.

VIII. DISCUSSION

A. Strengths

- **Model interpretability.** Prediction can be incorporated into biologists' workflows because predictions are based on 64 unique biologically significant features (e.g. amino acid composition) and are therefore explainable to domain experts.
- **Robust to class imbalance.** Features like `scale_pos_weight` and `class_weight` allow us to adjust model parameters to ensure balance during training and prevent overwhelming classification.
- **Rapid prototyping.** Non-programmers can simply input an NCBI accession ID and receive predictions and confidence scores within seconds.
- **Near-perfect Latex allergen detection.** Latex allergens are separate from the other classes of allergens. As a result of the unique `aa_E` signature, Latex class prediction has the highest accuracy (30/31 correct in the test set).

B. Limitations

- **Latex class size.** There are fewer unique Latex allergens on NCBI than other allergens, potentially leading to under-generalisation of rare Latex variants.
- **Dust Mite confusion.** 11–13 Dust Mite allergens have been misclassified as Pollen or Venom. This is a real-world

biological problem and future iterations can add features at the subfamily level or use constructs like protein domain annotation.

- **Novel folds.** Models classify based on sequence. Proteins with entirely novel folds but low sequence allergenicity signal may be missed. Incorporating structural data (e.g. from AlphaFold2) would allow for even further classification.
- **NCBI dependency.** For real-time prediction, the NCBI Entrez API must be accessed. Sequence retrieval can be slow for busy servers or large proteins.

C. Comparison with Existing Tools

AllerTop 2.0 (k-nearest neighbour on amino acid composition), AllerHunter (SVM with physicochemical properties), and iALLERGEN (ensemble models) are examples of state-of-the-art allergenicity predictors. Our Stage A XGBoost model achieves an ROC-AUC of 0.994, which is comparable to or better than published benchmarks for similar binary allergen classification tasks. Most approaches do not predict allergen category, and thus our Stage B type classifier is novel.

IX. CONCLUSION

In this study, we implemented a two-stage machine learning pipeline for protein allergen detection and allergen type classification. Over 30,000 protein sequences were gathered from the NCBI. We defined 64 physicochemical and dipeptide features, and utilised XGBoost (Stage A) and Random Forest (Stage B) classifiers with extensive validation.

The final system achieves:

- **Stage A:** 96.1% accuracy, ROC-AUC 0.994, Average Precision 0.993 on the held-out test set.
- **Stage B:** 94.4% macro-F1 for six-way allergen type classification on allergen-only rows.

The system is live and fully deployed via a web application that can be accessed from any browser. It functions as a rapid, first-pass screening tool that captures sequences from the NCBI in real time. The combined system is useful in the realm of food safety, health, and environmental protection. Future research will focus on the use of structural features, dander allergens from animals, and protein language models based on transformers as alternative feature sources.

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